2. **Vaccines:** Two types of oral vaccines have been tried recently:
   - Killed oral whole cell vaccines
   - Live oral vaccines

### 7.10.8 Treatment

1. Oral Rehydrationtherapy: The severe dehydration & salt depletion can be treated by oral rehydration therapy (as recommended by WHO).
2. Antibiotics: It is of secondary importance, oral tetracycline was recommended for reducing the period of Vibrio excretion.




An ideal cholera vaccine is yet to be found.

## 7.11 *Mycobacterium Tuberculosis* (Tubercle Bacillus)

The genus *Mycobacterium* is distinguished by its thick, complex, lipidrich waxy cell walls. This high lipid content (Mycolic acids) imparts the characteristic of acid fastness or resistance to decolorizationby a strong acid after staining with carbol fuchsin. Many of the Mycobacterial species are saprophytes but several species are highly significant human pathogens. *Mycobacterium tuberculosis* is the causative agent of tuberculosis (TB) It is a killer disease and ranks as one of the most serious infection diseases of the developing countries. TB is primarily a disease of the lungs but may spread to other sites of the body.

The name *Mycobacterium tuberculosis* is derived form,

- Mycobacterium (Greek) – Fungus like bacterium
- Tuberculosis (Latin) – Swelling or Knob

### 7.11.1 Morphology

They are acid fast bacilli, slightly curved rods, it may occur singly or in small clumps. They are non–motile, non–sporing, and non-capsulated.

### 7.11.2 Cultural Characteristics

They are obligate aerobe, optimum temperature is 37°C and optimum pH is 6.4–7.0. The pathogen grows on an enriched culture media – Lowenstein Jensen medium. The colonies appear in about 2–3 weeks. The colonies are dry, rough, raised, irregular colonies with a wrinkled surface. Initially creamy white and becoming yellowish later (Figure 7.19).

### 7.11.3 Pathogenesis

Human tuberculosis is divisible into two form, they are Primary TB and Secondary

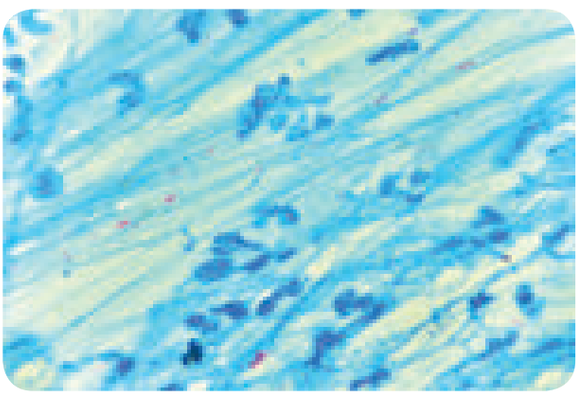


**Figure 7.19:** Acid fast staining of *Mycobacterium tuberculosis*



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